# Education

*Background in Theoretical Physics with Scientific Computing*

SKILLS

Maths | Physics | Computing (Python, Fortran)

01 · MSC · 2017 – 2018 · GPA 3.7

## Imperial College London

*Master of Science, Physics*

DISSERTATION

**Activation energies of ionic migration in perovskites, via Electrochemical Impedance Spectroscopy**

### Experimental physics + computation

Impedance measurements on perovskite solar cells, analysed and modelled in Python.

### Wrote an EIS fitting algorithm

Fit equivalent-circuit models to impedance data by minimising mean-square error.

### Results matched theory

Activation energies agreed with prior studies. Linear regression, Matplotlib visuals.

02 · BSC · 2011 – 2015 · GPA 3.66

## Universitas Gadjah Mada

*Bachelor of Science, Condensed Matter & Materials Physics*

UNDERGRADUATE THESIS

**Electronic Transport in Semiconducting Graphene**

### Theory + computation

Python (NumPy, Matplotlib), Fortran and MATLAB.

### Modelled a graphene pn junction

Two differently doped graphene layers, assessed for semiconductor devices.

### Lengthy analytical derivations

Worked the maths through by hand, then solved and visualised it numerically.

**Physics taught me to model messy data, then write the code that fits it.**

*London · Yogyakarta*